

Ali Vaziri

Machine Learning, Simulation, Control, and Estimation for Physical AI
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I. Professional Summary

AI/ML engineer and Ph.D. researcher building learning, planning, and control systems for autonomous vehicles, robotics, and physical AI more broadly. My work spans multimodal trajectory prediction, large-scale simulation, physics-guided learning, Bayesian inference, and model predictive control, backed by distributed PyTorch training at scale. At General Motors' Cruise team, I built large-scale data, training, and evaluation pipelines for autonomous-driving simulation. At Mitsubishi Electric Research Labs, I developed physics-informed neural models that combine continuous-time dynamics with physical constraints to deliver accurate predictions from limited real-world data. My graduate research has produced control and estimation methods that run up to 200× faster than standard benchmarks, published in *IEEE Transactions on Robotics*, *Applied Thermal Engineering*, IEEE CDC, and ACC (see Selected Publications, below).

II. Industry Research Experience

Summer 2026 Sunnyvale, CA	AI/ML Software Engineering Intern, General Motors – Cruise Team Scalable ML-Based Motion Reconstruction for Autonomous-Vehicle Simulation - Built an end-to-end multimodal ML system that reconstructs missing vehicles' motion in recorded-driving simulations, enabling more complete and realistic testing of autonomous-driving behavior. - Built cloud data and featurization pipelines for more than 38,000 highway scenes (approx. 1 TB of raw tracking data) using Google Cloud Storage, Dataflow, and memory-mapped feature stores. - Scaled training across 8 H100 GPUs with PyTorch DistributedDataParallel, NCCL, and mixed-precision computation, cutting wall-clock training time from 12 hours to 3 hours. - On a held-out evaluation of approximately 58,000 vehicle trajectories, reduced best-candidate average position error by 24% relative to the team's prior baseline. - Reworked full-test evaluation around streaming data, parallel loading, and memory-mapped features, cutting runtime from 60+ minutes to about 9 minutes and enabling faster model comparisons and data-quality studies. - Built a scenario-level collision-detection metric that checks oriented vehicle footprints at 10 Hz, removes duplicate collision pairs, and complements trajectory accuracy with collision, smoothness, and timing measurements.
Summer 2024 Cambridge, MA	Research Intern, Mitsubishi Electric Research Laboratories (MERL) Machine Learning Modeling and Optimization - Developed a physics-constrained Neural ODE–GRU model for vapor-compression HVAC systems, improving forecasting accuracy by 14% and runtime by 5.7× relative to the prior modeling approach. - Combined continuous-time neural dynamics, recurrent sequence modeling, and known physical constraints to obtain accurate ML predictions from limited data. - Delivered reusable PyTorch software covering the full pipeline: data processing, model training, evaluation, and GPU-accelerated control experiments.

III. Education

Ph.D.	Mechanical Engineering – University of Kansas 2023–2025 Transferred to Michigan State University 2026–present Research focus: Optimal inferential control of complex, large-scale systems
M.S.	Mechanical Engineering – University of Kansas 2023–2025 Earned en route to Ph.D. – GPA: 4.00/4.00

IV. Academic Research Experience

- 2023–present **Graduate Research Assistant** — University of Kansas (2023–2025); Michigan State University (2026–present)
Scientific Machine Learning and Intelligent Control/Estimation
- Developed a new inferential framework for efficient, fast optimal-control methods such as model predictive control (MPC), applied to neural-network-based vehicle models, high-dimensional fluids, and thermofluid systems; tested configurations ran up to **200× faster** than CasADi/IPOPT and MPPI benchmarks.
 - Developed continuous-time optimal control of Neural ODEs for soft-robot manipulators.
 - Developed GPU-compatible tensor-variate ensemble Kalman smoothing for high-dimensional spatial models, achieving up to **18.6× faster computation** and roughly 11× lower memory use than a vectorized GPU baseline.
 - Designed robust motion planning with Student's- t distributions, sequential Monte Carlo, and ensemble Kalman smoothing to handle non-Gaussian uncertainty and outliers.
 - Trained GRU and ResNet vehicle-dynamics models on real-world driving data and connected them to Bayesian planning and conventional trajectory optimization pipelines.

V. Selected Publications

- I. Askari, **A. Vaziri**, X. Tu, S. Zeng, and H. Fang, “Model Predictive Inferential Control of Neural State-Space Models for Autonomous Vehicle Motion Planning,” *IEEE Transactions on Robotics*, vol. 41, pp. 3202–3222, 2025. [Paper]
- **A. Vaziri** et al., “Modular Deep Learning Framework for Vapor-Compression HVAC Systems: Scalable, Efficient, and Physically Consistent Modeling,” *Applied Thermal Engineering*, 2026. [Paper]
- **A. Vaziri** et al., “Continuous-Time Optimal Control of Neural ODEs via Bayesian Inference,” *IEEE CDC*, 2026.
- **A. Vaziri**, et al., “Bayesian Inferential Motion Planning Using Heavy-Tailed Distributions,” *ACC*, 2025. [Paper]
- **A. Vaziri** et al., “Optimal Inferential Control of Convolutional Neural Networks,” *ACC*, 2025. [Paper]

VI. Technical Skills

Machine Learning and Modeling: Deep learning, generative modeling, multimodal learning, sequence and time-series modeling, convolutional neural networks (CNNs), recurrent neural networks, Neural ODEs, probabilistic modeling, physics-guided learning, scientific machine learning, system identification

Control, Planning, and Estimation: Model predictive control, Bayesian inference, trajectory optimization, robust motion planning, Kalman filtering and smoothing (EKF, UKF, EnKF/EnKS), particle filtering, sequential Monte Carlo, uncertainty-aware planning

Programming and ML Systems: Python, C/C++, CUDA, PyTorch, TensorFlow, NumPy, SciPy, DistributedDataParallel (DDP), NCCL, mixed-precision training, distributed GPU computing, performance profiling, unit and integration testing

Simulation and Physical AI: Road-to-Sim, Webviz, scenario conversion, world/scene reconstruction, simulation testing, trajectory evaluation, collision and quality metrics

Robotics and Optimization Tools: ROS, MuJoCo, PyBullet, CasADi, IPOPT, MATLAB/Simulink, SolidWorks

Cloud, Data, and Development Tools: Google Cloud Storage, Dataflow, RoboFlow, memory-mapped datasets, Bazel, Linux, Git